Problem.

Find all complex solutions of the following equation, together with their multiplicities:

$$1 + x + x^2 + x^4 + x^5 + x^6 + x^8 + x^9 + x^{10} = 0. \tag{1}$$

Write $P(x)$ for the polynomial on the left-hand side. Since P has degree ten, the fundamental theorem of algebra guarantees exactly ten complex roots when multiplicities are counted. The problem is to determine where these roots lie and which of them are repeated. A conspicuous feature is the absence of the powers x^3 and x^7; we shall first exploit a simple factorization suggested by the remaining terms.

An elementary ansatz.

The exponents appear in three consecutive groups, separated by gaps of one missing power. Before developing any theory, group the terms accordingly:

$$P(x) = (1 + x + x^2) + (x^4 + x^5 + x^6) + (x^8 + x^9 + x^{10}). \tag{2}$$

Each group is a multiple of the same quadratic polynomial. Pulling out these powers of x immediately yields

$$\begin{aligned} P(x) &= (1 + x + x^2) + x^4(1 + x + x^2) + x^8(1 + x + x^2) \\ &= (1 + x + x^2)(1 + x^4 + x^8). \end{aligned} \tag{3}$$

Thus a degree-ten equation has been reduced to two elementary factors. The manipulation is straightforward; the less obvious question is why the original exponents were organized in precisely this way.

Solving the two factors.

The first factor is quadratic. Its two roots are the primitive cube roots of unity, denoted by ω and ω^2:

$$1 + x + x^2 = 0 \quad \Longrightarrow \quad x = \frac{-1 \pm i\sqrt{3}}{2} \in \{\omega, \omega^2\}, \qquad \omega = e^{2\pi i/3}. \tag{4}$$

For the second factor, make the equally elementary substitution $t = x^4$. The same quadratic equation now gives

$$1 + t + t^2 = 0 \quad \Longrightarrow \quad x^4 = \omega \text{ or } x^4 = \omega^2. \tag{5}$$

The fourth roots can be written explicitly in polar form:

$$x = e^{i\theta}, \qquad \theta = \pm\frac{\pi}{6} + \frac{k\pi}{2}, \qquad k = 0, 1, 2, 3. \tag{6}$$

The second factor therefore supplies eight distinct roots on the unit circle.

The repeated roots appear.

At first sight the first factor seems to contribute two additional roots. Yet its roots already occur among those of the second factor. Indeed, $\omega^3 = 1$ implies

$$\omega^4 = \omega, \qquad \omega^8 = \omega^2, \qquad 1 + \omega^4 + \omega^8 = 1 + \omega + \omega^2 = 0. \tag{7}$$

The same calculation holds with ω replaced by ω^2. Consequently, both factors vanish at ω and ω^2. These are double roots, while the other six roots of the second factor are simple. The multiplicities give the required count:

$$2 \cdot 2 + 6 \cdot 1 = 10 = \deg P. \tag{8}$$

We have already solved the problem by direct algebra. The remaining question is conceptual: why do the roots form such a regular pattern, and how can the factorization be recognized systematically rather than guessed?

The structure of the exponents.

Every exponent of P is uniquely $r + 4s$, where $r, s \in \{0, 1, 2\}$:

	$r = 0$	$r = 1$	$r = 2$
$s = 0$	0	1	2
$s = 1$	4	5	6
$s = 2$	8	9	10

In base 4 these are the nine two-digit numerals made of 0, 1 and 2. The missing exponents, $3 = (03)_4$ and $7 = (13)_4$, would require the digit 3. The ansatz in Eq. (3) now follows systematically:

$$\begin{aligned} P(x) &= \sum_{s=0}^{2} \sum_{r=0}^{2} x^{r+4s} = \left(\sum_{r=0}^{2} x^{r} \right) \left(\sum_{s=0}^{2} x^{4s} \right) \\ &= (1 + x + x^2)(1 + x^4 + x^8). \end{aligned} \tag{9}$$

A closed form.

Each factor in Eq. (3) is a finite geometric series. The two corresponding identities are

$$(1 - x)(1 + x + x^2) = 1 - x^3, \tag{10a}$$

$$(1 - x^4)(1 + x^4 + x^8) = 1 - x^{12}. \tag{10b}$$

Multiplying them gives a polynomial identity valid for every $x \in \mathbb{C}$:

$$(1 - x)(1 - x^4)P(x) = (1 - x^3)(1 - x^{12}). \tag{11}$$

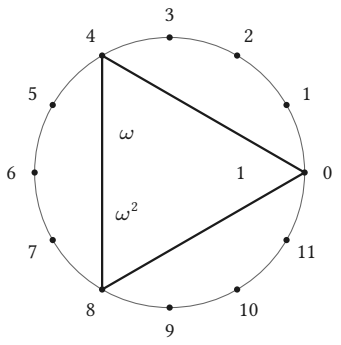
Every root of P is therefore a root of the product $(1 - x^3)(1 - x^{12})$, hence a cube root or a twelfth root of unity. Since 3 divides 12, every cube root of unity is already a twelfth root of unity. Equation (11) thus places all the zeros of P among twelve equally spaced points of the complex unit circle, although it remains necessary to determine which of those points survive and with what multiplicities.

Roots of unity.

For any positive integer n, the equation $z^n = 1$ has exactly n solutions,

$$\begin{cases} z_k = e^{2\pi ik/n} = \cos\dfrac{2\pi k}{n} + i\sin\dfrac{2\pi k}{n}, \\ k = 0, 1, \ldots, n-1. \end{cases} \tag{12}$$

These points are the vertices of a regular n-gon inscribed in the unit circle, with one vertex at $z = 1$. Each is a simple root of $z^n - 1$, because the derivative nz^{n-1} does not vanish there. For $n = 12$ the arguments are $k\pi/6$, and the cube roots of unity correspond to $k = 0, 4, 8$.



Vertices indexed by k; triangle: cube roots.

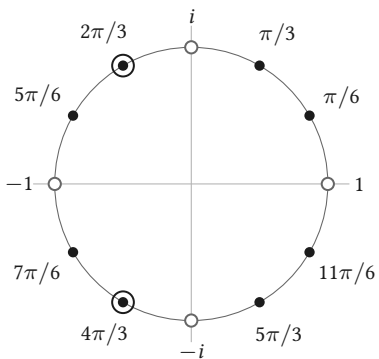
Counting multiplicities.

At each twelfth root $\zeta_k = e^{ik\pi/6}$, compare the multiplicities on both sides of Eq. (11). Multiplicities add under multiplication:

		Multiplicity in		
Root	k	$(1-x^3)(1-x^{12})$	$(1-x)(1-x^4)$	$P(x)$
1	0	2	2	0
$i, -1, -i$	3, 6, 9	1	1	0
ω, ω^2	4, 8	2	0	**2**
The other six	1, 2, 5, 7, 10, 11	1	0	1

The roots ω and ω^2 are double, the other six are simple, and 1, i, -1 and $-i$ are excluded. This accounts for all ten roots, counted with multiplicity.

The roots on the unit circle.



Writing $x = e^{i\theta}$, the six simple roots occur at

$$\theta \in \left\{ \frac{\pi}{6}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{5\pi}{3}, \frac{11\pi}{6} \right\}. \tag{13}$$

The two double roots occur at

$$\theta \in \left\{ \frac{2\pi}{3}, \frac{4\pi}{3} \right\}, \qquad x \in \{\omega, \omega^2\}. \tag{14}$$

● Simple; ⦿ double; ○ not a root. Since P has real coefficients, all non-real roots occur in conjugate pairs with equal multiplicity, as the symmetry of the diagram confirms.

Check: the substitution $t = x^4$.

The substitution $t = x^4$ provides an independent check. Solving its quadratic equation gives

$$t = \frac{-1 \pm i\sqrt{3}}{2} = e^{\pm 2\pi i/3}. \tag{15}$$

Taking the fourth roots of $x^4 = t$ produces the two angular families

$$\theta = \pm\frac{\pi}{6} + \frac{k\pi}{2}, \qquad k = 0, 1, 2, 3, \tag{16}$$

which, after reduction modulo 2π, become

$$\begin{aligned} \theta_+ &\in \left\{ \frac{\pi}{6}, \frac{2\pi}{3}, \frac{7\pi}{6}, \frac{5\pi}{3} \right\}, \\ \theta_- &\in \left\{ \frac{11\pi}{6}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{4\pi}{3} \right\}. \end{aligned} \tag{17}$$

The roots ω and ω^2 occur in both factors and are therefore double.

Factorization over the real numbers.

Since P has real coefficients, every non-real zero $e^{i\theta}$ is accompanied by its complex conjugate $e^{-i\theta}$. The pair determines a quadratic polynomial with real coefficients:

$$(x - e^{i\theta})(x - e^{-i\theta}) = x^2 - 2x\cos\theta + 1. \tag{18}$$

The four conjugate pairs have angles $2\pi/3$, $\pi/3$, $\pi/6$, and $5\pi/6$. Including the multiplicity of the cube roots gives

$$\begin{aligned} P(x) = (x^2 + x + 1)^2(x^2 - x + 1) \\ \times (x^2 - \sqrt{3}\,x + 1)(x^2 + \sqrt{3}\,x + 1). \end{aligned} \tag{19}$$

Each quadratic factor has discriminant $4\cos^2\theta - 4 = -4\sin^2\theta < 0$, because none of the four angles equals 0 or π. Thus the displayed factors are irreducible over $\mathbb{R}$ and P has no real zeros.

Factorization over the rational numbers.

The two factors containing $\sqrt{3}$ have real coefficients but do not individually have rational coefficients. Their product eliminates the irrational terms:

$$(x^2 - \sqrt{3}\,x + 1)(x^2 + \sqrt{3}\,x + 1) = x^4 - x^2 + 1. \tag{20}$$

Consequently, the factorization in Eq. (19) can be rewritten entirely with rational coefficients:

$$P(x) = (x^2 + x + 1)^2(x^2 - x + 1)(x^4 - x^2 + 1). \tag{21}$$

Each factor has leading coefficient 1: the coefficient multiplying its highest power of x is unity. A polynomial with this property is called *monic*. The factorization above also makes clear that passing from real to rational coefficients changes the grouping of roots into factors, without changing the roots of P or their multiplicities.

Monic and cyclotomic polynomials.

Recall that a polynomial is *monic* if its leading coefficient is 1; for example, $x^2 + x + 1$ is monic, whereas $2x^2 + x + 1$ is not. An n-th root of unity is *primitive* if its multiplicative order is exactly n, meaning that no smaller positive power equals 1. The *cyclotomic polynomial* $\Phi_n(x)$ is the monic polynomial whose zeros are precisely the primitive n-th roots of unity.

For the orders appearing in this problem, these polynomials are

$$\begin{aligned} \Phi_3(x) &= x^2 + x + 1, \\ \Phi_6(x) &= x^2 - x + 1, \\ \Phi_{12}(x) &= x^4 - x^2 + 1. \end{aligned} \tag{22}$$

The rational factorization therefore has the concise form

$$P(x) = \Phi_3(x)^2 \, \Phi_6(x) \, \Phi_{12}(x). \tag{23}$$

The squared factor Φ_3 records the double multiplicity of the primitive cube roots of unity.

Conclusion.

The solutions of Eq. (1), with their multiplicities, are

$$x = e^{ik\pi/6}, \qquad k \in \{1, 2, 4, 5, 7, 8, 10, 11\}, \qquad k = 4, 8 \text{ double.} \tag{24}$$

They are the twelfth roots of unity other than ± 1 and $\pm i$, with ω and ω^2 occurring twice. No root is real. The elementary grouping in Eq. (3) proves the factorization immediately, while the base-4 explanation and the geometric-series identity reveal the regularity behind that algebraic observation and justify the complete multiplicity count.

The construction extends beyond the present example. If $m, b \geq 2$ and $\gcd(m, b) = 1$, every m-th root of unity other than 1 is a double root of

$$\left(\sum_{r=0}^{m-1} x^r \right) \left(\sum_{s=0}^{m-1} x^{bs} \right). \tag{25}$$

The problem studied here corresponds to $m = 3$ and $b = 4$.

References.

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